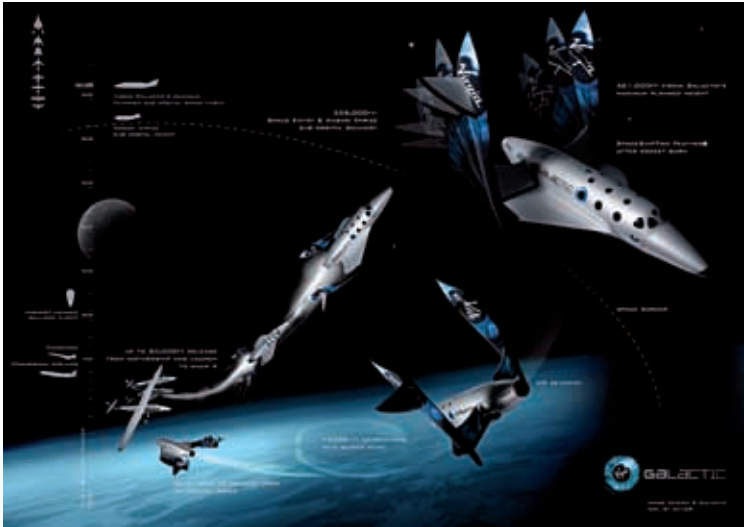




The detailed flight of the SpaceShipTwo

very high drag as the spacecraft descends through the upper regions of the atmosphere. The feather configuration is also highly stable, effectively giving the pilot a hands-free re-entry capability, something that has not been possible on spacecraft before, without resorting to computer controlled fly-by-wire systems. The combination of high drag and low weight (due to the very light materials used to construct the vehicle) mean that the skin temperature during re-entry stays very low compared to previous manned spacecraft and thermal protection systems such as heat shields or tiles are not needed. During a full sub-orbital spaceflight, at around 70,000 feet following re-entry, the feather lowers to its original configuration and the spaceship becomes a glider for the flight back to the spaceport runway.

WhiteKnightTwo, or Eve, is the mothership and launch platform for SpaceShipTwo. Due to the demanding design requirements of this vehicle, WhiteKnightTwo has become the world's largest all carbon composite aircraft with an amazing 140 foot wing span. Powered by four Pratt and Whitney PW308A engines, it has a unique heavy lift, high-altitude capability and an open architecture design which provides for maximum versatility in the weight, mass, and volume of its payload. It has the power, strength and manoeuvrability to provide for pre space-flight, positive G force and zero G astronaut training as well as a lift capability which is over 30 per cent greater than that represented by a fully-crewed SpaceShipTwo.